

Improving Agent-Based Route Predictions for As-The-Crow-Flies Navigation

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ABSTRACT

Mobile navigation has become a ubiquitous technology through mobile devices and smartphones. One of these devices' most-used navigation methods is the turn-by-turn method, first used in car-based navigation systems. However, pedestrians and cyclists increasingly use other methods such as the as-the-crow-flies (ATCF) navigation method instead of the classical turn-by-turn navigation. Instead of giving fixed instruction, ATCF navigation only indicates the straight-line direction to the destination and users have to make their own decision when and where to turn. In this paper, we improve the route predictions for such alternative navigation methods as ATCF by incorporating a model on users' angle estimation error. We show that the route predictions are closer to actual human navigation behavior than methods proposed in related work.

CCS CONCEPTS

• **Human-centered computing** → *Empirical studies in HCI*; *User studies*; *Mobile devices*.

KEYWORDS

navigation, agent, simulation, as-the-crow-flies, routing

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1 INTRODUCTION

Many people rely on mobile navigation apps for their everyday commute, leisure trips, and generally getting from A to B. For car drivers, cyclists, and pedestrians, these apps often use turn-by-turn (TBT) instructions of in-car navigation systems and navigation apps on smartphones or smartwatches. However, in the field of pedestrian and cyclist navigation, the trend is moving away from classical turn-by-turn navigation and towards other navigation methods. One example is the as-the-crow-flies (ATCF) navigation method (see Figure 1). The ATCF navigation method provides users

with a constant indication of the direction to the destination. Given such information, users will choose the street with the least angle deviation to the destination at decision points. While being found suitable in many use cases [1, 14, 17] and providing an alternative navigation and user experience, the ATCF navigation method also has disadvantages compared to the TBT method. These problems include users ending up in dead ends [15, 17], going in the "wrong" direction [13, 14, 17], and not knowing when to take a turn [1, 17]. To prevent such errors, mobile ATCF navigation systems need to better predict and anticipate their users' potential routes and navigation behavior. This poses a challenge as there is no fixed pre-computed route as, for example, there is with turn-by-turn navigation. Existing methods to predict these routes use heuristics like the least angle strategy (LAS), which describes how humans navigate given only the direction to the destination [10]. However, existing implementations of this heuristic show a mismatch between the heuristic's prediction and actual human behavior [17]. One reason for this mismatch is the task of actually choosing the least angle at any decision point. User studies with ATCF navigation have shown that this decision can be very difficult [1, 17]. While an agent will always choose the better angle, no matter how small the deviation is, a human might not even realize the difference between a 20° and 23° angle and potentially choose a sub-optimal path. While related work has successfully explored software-agent-based simulation based on the LAS [10], this work extends existing approaches by creating and including a model for this angle estimation error into the agent's decision process, making its performance closer to that of humans. In the context of this work, we test a novel uncertainty (UC) agent against an existing LA agent. We compared the routes of our novel agent with route data of humans using the ATCF navigation. The results show that our novel agent better represents human navigation behavior under ATCF navigation guidance than existing LAS approaches. With this work, we contribute a model that simulates a human error when judging angles at decision points. Furthermore, we extend existing agents and thus improve potential agent-based modeling of human navigation behavior.

2 RELATED WORK

This section presents insights into how humans perceive angles in navigation, how this perception affects their behavior, and discuss existing approaches of agent-based modeling of ATCF navigation behavior and applications of the navigation methods in related studies.

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Figure 1: A comparison of visual as-the-crow-flies and turn-by-turn navigation methods for a typical navigation situation.

2.1 Perception of Angles by Humans

People in unknown environments need three essential components to find a destination: their location, orientation, and the metric properties of the environment (e.g. how many kilometers am I away from my destination) [6]. Based on these components, people form a cognitive map of their environment and use it to find their way around [6]. To do that, humans rely on their perception for wayfinding and navigation [8]. The environment is perceived and processed visually, which can lead to misjudging distances and distorting the cognitive map of the environment [8]. This results in navigation and wayfinding being shaped differently by the underlying individual model. This was already suggested in 1992, when it was shown in a small experimental setup that people, even with good local knowledge, are not able to locate places using distances or angles [7]. In order to compensate for these inaccuracies when navigating through an unknown environment, humans employ different strategies. Dalton [5] observed that they strive to choose the least possible angular deviation from their current path on the way to their destination, i.e., to walk straight ahead for as long as possible. This preference is consistent with the experiments of Bailenson et al. [2] conducted in 2000. Here it was shown that people chose routes that contained the longest possible straight sections. If several routes had an equal proportion of straight sections, those were preferred which contained the least number of turns at the beginning of the route [2]. This strategy is also known as "longest leg first".

When it comes to the perception of angles, Sadalla and Montello suggested in 1989 that humans are very good at estimating right angles and probably diagonal angles (45°, 135°, 225°, and 315°) because the Earth's gravitational field provides a fixed point [16]. An experiment showed that humans could estimate angles close to 90° well, while angles between 0° and 90° are overestimated and angles between 90° and 180° are underestimated. This suggests that human perception tends to bias angles toward 90° [16]. In 1993, Loomis et al. conducted further experiments on the precision of human estimated angles confirming the findings of Sadalla and Montello [11]. The studies above suggest that perceiving angles as part of a navigation method could cause some inaccuracies for certain angles. How exactly these inaccuracies affect ATCF navigation has been found by the following two studies. Both Albrecht

et. al. [1] and Savino et al. [17] tested how well participants were able to navigate given ATCF navigation instructions. However, the studies differed in the modality used to communicate the direction to the destination. While Albrecht et al. [1] used spatial audio, Savino et al. [17] used a visual interface. Still, both studies found that participants had difficulties making the right call for minor angular deviations between two streets.

In this paper, we are quantifying this difficulty with the help of a pre-study in which participants have to make a call between two streets given ATCF instructions with minor angular deviations. This specific task is essential for ATCF navigation because the correct interpretation of angles is central to successfully using the navigation method. A better understanding of this error will help us develop a better agent-based model to predict human wayfinding with the ATCF navigation method.

2.2 Modeling Human Wayfinding with ATCF Navigation

The most suitable model to describe people's navigation behavior with ATCF navigation is the least angle strategy proposed by Hochmair and Frank in 2000 [9]. Here, a person selects the street with the least angle deviation from the direct target direction at each turn when navigating to a visible location like a tall building or other landmark [9]. To evaluate the LAS, Hochmair [10] conducted a virtual simulation. In order to build an agent that was able to traverse a street network with close to human-like performance, he presents the memorized least angle strategy as the basis for his agent. In addition to the above-described behavior, the agent could memorize the last node visited but could not return to this last node in a subsequent step (except when in a dead end) [10]. In this paper, we will use this implementation as the basis for implementing our own agents.

While the LAS is a heuristic describing human wayfinding behavior, the ATCF navigation method utilizes the LAS by providing the direction to the destination throughout the navigation task for better orientation and fewer estimation errors. However, field studies have shown that the ATCF method has a few drawbacks. A recent study by Savino et al. [17] on ATCF navigation found that cyclists had problems going into a direction that deviated from the indicated direction to the destination, similar to what Pielot



(a)



(b)

Figure 2: Two of the five intersections which participants saw in the online survey. (a) Shows an intersection without a navigation aid, where participants had to choose the street that would lead them to the destination, which was "exactly straight ahead". (b) Shows an intersection with the instruction angle pointing to the right.

et al. [13] found for pedestrians. Furthermore, participants were unsure when to take a turn due to the lack of a map, similar to the results of Albrecht et al. [1]. Finally, participants ran into dead ends while exploring possible routes to the destination, as also seen in the study of Robinson et al. [15].

In order to anticipate and circumvent these problems, we need to predict someone's path when using ATCF navigation successfully. Since there is no pre-computed path, we have to closely model human decision-making to calculate a possible path they might take to be able to react to problems they might run into. Based on the previously described difficulty to assess small angle deviations, in this work, we will investigate whether an agent using the LAS together with a human-like angular estimation error can represent the navigation behavior of participants using the ATCF navigation method [17] better than existing implementations from the literature [10].

3 PRE-STUDY

In this pre-study, we examine the difficulty of choosing the turn with the least angle deviation to the destination, as suggested by the LAS. The study explores how well humans can make navigation decisions based on estimating small angles of less than 15° , possibly identifying other influencing factors, and deriving a function from the collected data that models human decisions uncertainty in this task.

3.1 Study Design

Five traffic intersections were photographed to carry out the study, each with two possible turns. The perspective of the photos was chosen, as best as possible, from the apex of the two outgoing roads so that the subject's point of view would not later dictate any tendency about the direction (see Figure 2). In an online survey, participants were asked to decide which street they would take, given the information shown on the smartphone. The smartphone showed an ATCF navigation interface with a compass needle pointing towards the destination. Whenever there was no smartphone, the instruction was that the destination was precisely straight ahead. The angles shown on the phone were: 0° (straight ahead), 2.5° , 5.0° , 7.5° , and 15° . All angles greater than zero were once pointing to the

right, and the left with each road intersection. Combined with the "no smartphone" situation, this made for ten different situations per intersection. Based on five intersections, this results in a combination of 50 different situations. In total, 64 respondents participated in the study via an online survey. The study itself was conducted in English. Participants ranged in age from 18 to 54, with an average age of 27 ($SD=7.67$). The participant pool consisted of 32 females, 30 males, and two non-binary participants. The 50 situations were divided into five groups, each with ten different situations to keep the survey time short while still guaranteeing an even distribution of results for each configuration of road intersections and associated angles. Each of these groups includes each of the available angles, once pointing to the left and once to the right and an image without instructions in randomized order. During the survey, each participant evaluated two groups (20 situations).

3.2 Results

Based on participants' responses, we calculated the percentage of correct decisions per angle and direction. Figure 3 shows the percentage of the correct decisions broken down by angle and direction, so the quantities include all underlying intersections in equal amounts. For the instructions oriented to the right, at 2.5° a total of 77.95 % of the responses were correct, at 5.0° we saw 86.82 % correct responses, at 7.5° 77.78 % of the responses were correct, and at 15° a total of 89.84 % were correct. For the left-facing instructions, at 2.5° 61.42 % of the responses were correct, the instructions at 5.0° again had more correct responses at 68.50 % than 7.5° at 66.41 %, while the 15° angle had 76.38 %. Based on the collected data, we used linear regression to construct an uncertainty function that maps human error to decisions with small angles. For this purpose, the data of the angles from Figure 3a were combined into angle classes in Figure 3b. From this, the following function can be derived: $f(x) = 0.009319 \cdot x + 0.686236$.

In the following section, we will extend the existing LAS agent from Hochmair [10] with the derived uncertainty function to introduce an error in the assessment of the angles for the decision making of the agent. We will compare this new agent to the existing agent to test which one achieves a higher overlap of route decisions when compared to human navigators in an ATCF navigation task.

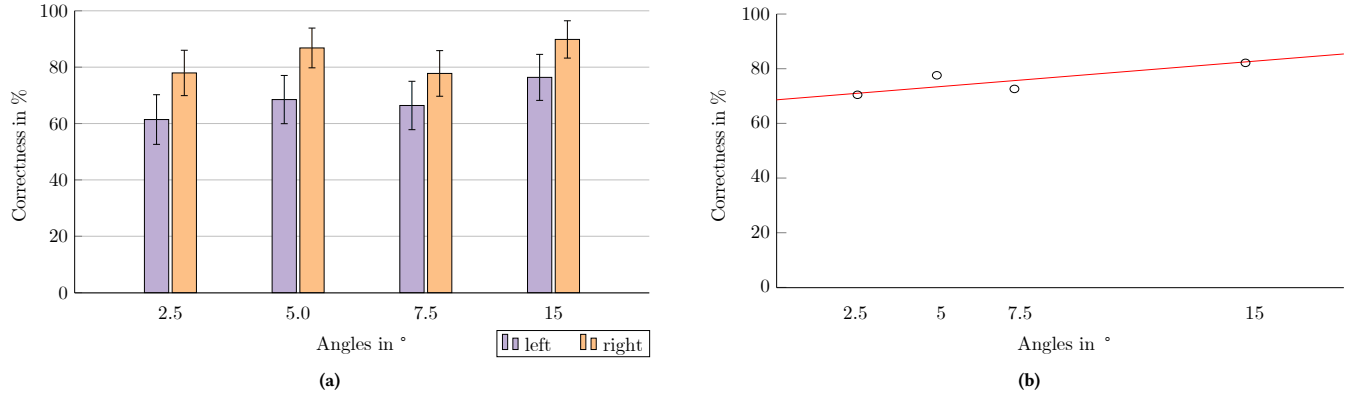


Figure 3: The correctness of participants judging the angles on the pictures. (a) Scores for the different angles point left and right. (b) The uncertainty function which we derived from the data.

4 SIMULATION & AGENTS

To evaluate how the uncertainty function influences the least angle agent and to test whether it makes its decisions more human-like, we create a simulation environment similar to that of Hochmair [10]. In this section, we will introduce OSMnx [3] for the simulation environment and our custom LA agents.

4.1 Simulation Environment

The basis of the least angle simulation is provided by OSMnx by Geoff Boeing [3, 4]. OSMnx provides the functionality to create and analyze directed graphs from Open Street Map (OSM) data [12]. The directed graph itself consists of nodes and edges with information about their latitude, longitude, the number of connected edges and other road traffic specific meta information. OSMnx provides information about the entry and exit degree of the node, incoming and outgoing edges, and information about traffic, road, and environment data. The edges have attributes about their start and destination node, the euclidean distance, as well as the bearing (compass direction), which describes the geographic orientation of an edge. This can be used to determine the angle between two edges or an edge and the target node. Our agent uses this functionality to decide which node to visit next at each decision point and will be affected by our uncertainty function.

4.2 Least Angle Agent & Uncertainty Agent

The following subsections describe the two agents we built. The first one is a replication of Hochmair's original agent [10] and the second one with the extended angle error implementation.

4.2.1 Least Angle Agent. The implementation of our least angle (LA) agent is based upon the implementation by Hochmair [10]. When finding its destination, the agent follows the following principle: The agent starts at an initial node, knowing its destination. Starting from its location, all further reachable options are considered in the form of nodes connected via edges. The direct predecessor node is then removed from this set, and the remaining nodes are sorted in ascending order according to their angular deviation

from the destination. The agent then travels to the node with the least angular deviation from the target that is still unvisited (has not been visited before). If all nodes have already been visited, the agent returns to the previous node of its location. This principle is repeated until the given destination is reached or one of the termination criteria is met. Termination occurs if the starting node has no neighbors or all available neighbors have already been visited, and the predecessor node has been visited more than once.

4.2.2 Uncertainty Agent. To examine and compare the findings from the preliminary study, we extend the above described LA agent. This new uncertainty (UC) agent should also use the least angle strategy as a basic structure. In addition, the findings from the study are integrated into the agent in the form of the described uncertainty function. This works with the help of the function from the regression model in Figure 3b, which is incorporated into the decision making of the uncertainty agent. The model describes

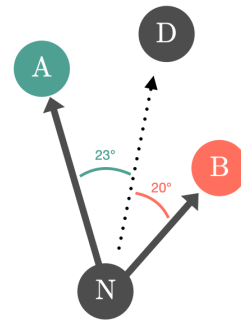


Figure 4: Example of an angle-decision situation for the agents. At node N with destination D, the agent can visit node A or B. Node B has a smaller angle deviation of 20°. The least angle agent will always choose to visit node B. However, the uncertainty agent will have a probability of 29% to choose node A and make a wrong decision.



Figure 5: The visualized overlap between the participant data from Savino et al. [17] and the respective agent. (a) Shows the result for experiment 1 and (b) the results of experiment 2 in Savino et al.'s [17] data.

the correctness of the decision, as a function of the angle difference, between two potential route options. Therefore, during the agent's pathfinding, we calculate the likelihood of the agent making a wrong decision for each decision between unvisited neighbours. Therefore, we take the difference of the angle deviation to the destination of the next two best neighbors and calculate the probability of error based on our error function. For example, if one option has an angle deviation of 20° to the destination and another 23° to the destination, resulting in a difference of 3° , the probability of making a correct decision would be 0.71%. Based on this probability, if the agent actually makes a mistake, the "worse" neighbor (in our example, the one with the 20° deviation) is chosen to be visited next. This example is depicted in Figure 4.

5 RESULTS

In this section, we will present the results of the comparison between the LA Agent and the UC Agent regarding the overlap of their routes with human performance. For this comparison, we use the data set from Savino et al.'s [17] study with a total of 13 participants across two experiments. To compare how similar the two agents performed against the human participants, we looked at the number of identical edges between participants and agents. In addition, a coverage factor of the routes is compared, which is calculated from the proportion of the route distance covered. Figure 5 shows the congruence of the route of the LA agent with the routes of the user data. To illustrate the scale of comparison, the subjects' route data are superimposed and drawn in red on the map. The parts of the route that were also traversed by the LA and UC agents are highlighted in yellow. While the LA agent behaves deterministic, coverage equality with the UC agent is not sufficient by simply comparing a single produced route. Instead, we produced a coverage graph from multiple routes of the UC agent. By comparing node pair by node pair from the study data with the data of the agents, we obtain coverage values that are shown in Table 1. Here, the UC agent covers 12.8% more edges in the first experiment and 15.5% more distance, respectively. In the second experiment, the UC agent covers 11.3% more edges and 12.8% more distance of the participants' paths. Overall, the built-in angle estimation error improved the coverage by 51 % on average compared to the

LA agent. Which means that the UC agent had a more human like performance than the existing LA agent.

6 DISCUSSION & CONCLUSION

Often humans are not the perfect machines we want them to be. Related work has shown that ATCF navigation sometimes relies on making very precise and, therefore, difficult angle estimations, resulting in bad navigation decisions [1, 17]. In some instances, a navigation decision that is obvious for a simple software agent is ambiguous for a human. In this work, we show (1) that there is a measurable effect of the difficulty to detect differences in small angles and (2) that this ability affects how humans use the navigation method. Subsequently, we learn that heuristics like the least angle strategy [9] are not sufficient to model this complex human behavior. Adding an angle estimation uncertainty lets the agent produce more similar routes to those of human participants in a navigation experiment than existing agents based on just the least angle strategy. While this result allows us to model human wayfinding with ATCF navigation better than before, our approach has some limitations that we discuss in the following paragraph.

The angle estimation error of our UC agent is based on the results of our pre-study. During the analysis of the results, we used the non-smartphone version of the decision task to check for biases towards left or right turns for each of the five intersections. We found that eventually, not only the compass direction had an effect on the decision process, but also other factors such as the direction cars are parked in, visible traffic signs, as well as the surface of the street (e.g. cobblestone versus asphalt). This led to an asymmetry in the results, shifting the majority of decisions in

	Experiment	Least Angle	Uncertainty
Overlap of edges in %	#1	24,6	37,4
	#2	20,0	31,3
Overlap of length in %	#1	34,2	49,7
	#2	25,0	37,8

Table 1: Overlap of the agent routes with the participant routes.

one or the other direction. While limiting the results, this gives us meaningful insights into potential aspects to further improve our agent. Furthermore, we limited our estimation task to a maximum of 15°. This means we can not know if our assumption of a linear function holds for larger degrees. However, we tested our setup initially with degrees up to 30° and found that participants correctly interpreted degrees larger than 15. Our results motivate further experiments with more than the one aspect that has been discussed in this work. Future work should consider these complex aspects and test them in a real-world environment.

In the context of this work, we explored the effect of the human angle estimation error for agent-based modeling of human wayfinding. For this purpose, we investigated human decision-making behavior, with compass instructions, at intersections in a pre-study and derived an angle estimation error function. This function was integrated into the decision making of a least angle agent. Simulations showed that the angle estimation error led to more human-like route choices than the existing LA heuristic. The UC agent model could be used in an ACTF+ [17] like interface to predict navigators' errors and reduce navigating into dead ends or taking non-target detours. This could help prevent user frustration and increase trust in the navigation device.

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